safety, and digital transformation.

5.5.4 Communication and technological applications

The integration of UAVs into communication networks marks a revolutionary step in extending global connectivity and supporting data-driven ecosystems. Fourth-generation drones function as airborne communication relays, mobile data centers, and IoT integration hubs.

- *Aerial communication relays*: Drones equipped with 5G/6G communication nodes provide temporary coverage in disaster zones or remote regions, ensuring continuity in communication and data services.
- *Satellite integration and network bridging*: UAVs act as intermediaries between satellite constellations and terrestrial networks, enhancing broadband connectivity and latency performance.
- *IoT and smart infrastructure support*: They facilitate real-time data collection from distributed sensors in smart cities, agriculture, and industrial zones, enabling automated decision-making through cloud–edge integration.
- *Cybersecurity and data protection*: With AI-based encryption and quantum-secure communication systems, drones protect sensitive military and civilian data transmissions against interference and cyber threats.
- *Emergency network restoration*: In post-disaster conditions, drones rapidly deploy as temporary communication towers, restoring connectivity and supporting rescue coordination.

These technological applications underline the growing role of fourth-generation UAVs as key enablers of global digital infrastructure, merging